BME 5910 - HW 1

Problems 1.9

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Aids:

- Textbook: Biomedical Optics: Principles and Imaging by Lihong Wang and Hsin-I Wu
- Data: https://omlc.org/spectra/hemoglobin/summary.html
- Copilot: Formatting data table

```
% Data
data = load("hemoglobin_data.mat").data;

lambda = data(:,1);
epsilon_ox = data(:,2);
epsilon_de = data(:,3);
```

```
% Plotting
plot(lambda, epsilon_ox, 'r-', 'DisplayName','HbO2')
hold on
plot(lambda, epsilon_de, 'b--', 'DisplayName','Hb')
hold off
yscale("log")
xlim([min(lambda), max(lambda)])
xlabel("Wavelength [nm]")
ylabel("Molar extinction coefficient [1/cm-M]")
title("Oxygenated and Deoxygenated Hemoglobin Extinction")
legend("Location","northeast")
```

